SMOLVLA / LIBERO | NINE EXECUTION RULES | 270 MATCHED EPISODES

How often should a robot replan?

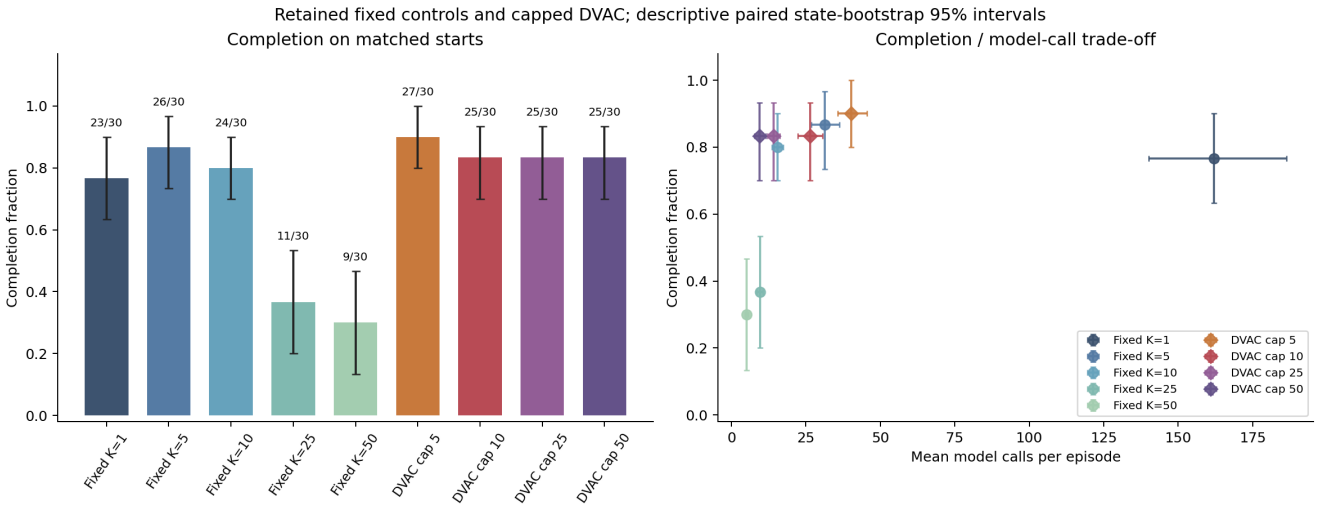
Fixed intervals and denoising-variance control with an execution cap

One pretrained model generates 50 future actions from images, robot state and an instruction. Five fixed intervals and four adaptive rules choose when to observe and replan. The generated horizon stays fixed; the executed prefix changes.

DVAC cap 10 completed 25/30 episodes; fixed K=10 completed 24/30 and DVAC cap 50 completed 25/30. All nine variants are reported. Completion and model calls are the common comparison measures.

Method	Completed	Completion % [95% interval]	Calls / episode [95% interval]	Calls / 100 actions
Fixed K=1	23/30	76.7 [63.3, 90.0]	162.0 [140.1, 186.5]	100.00
Fixed K=5	26/30	86.7 [73.3, 96.7]	31.3 [26.9, 36.3]	20.23
Fixed K=10	24/30	80.0 [70.0, 90.0]	15.5 [13.5, 17.4]	10.22
Fixed K=25	11/30	36.7 [20.0, 53.3]	9.6 [8.6, 10.6]	4.18
Fixed K=50	9/30	30.0 [13.3, 46.7]	5.1 [4.6, 5.6]	2.13
DVAC cap 5	27/30	90.0 [80.0, 100.0]	40.2 [35.6, 45.5]	29.24
DVAC cap 10	25/30	83.3 [70.0, 93.3]	26.4 [22.2, 30.7]	17.53
DVAC cap 25	25/30	83.3 [70.0, 93.3]	14.1 [12.1, 16.2]	8.97
DVAC cap 50	25/30	83.3 [70.0, 93.3]	9.5 [7.7, 11.5]	6.86

Intervals use a descriptive, task-stratified paired bootstrap. Every method uses the same 30 task/state settings and policy seed 11. See page 4 and the final page for interpretation.



Circles: fixed intervals; diamonds: adaptive caps. Error bars describe sampling variation over observed initial states. An apparent frontier of sample means is not proof of a superior method.

What changes: H, C and K

Symbol	Meaning	Setting
H	Future actions generated by one policy call	50 for every method
M	Internal Euler updates within that call	10 for every method
C	Maximum actions an adaptive call may execute	5, 10, 25 or 50
K	Actions selected for execution before a new observation	Fixed constant, or adaptive 1 through C

Flow matching inside one policy call

With the observation/instruction c held fixed, Gaussian noise X(0) becomes an action block. Internal iteration i is separate from future action index j. The simulator pauses during generation.

$$t_i = 1 - i/M,\ v_i = v_\theta(X^{(i)}, t_i, c)$$
$$X^{(i+1)} = X^{(i)} - v_i/M,\ i = 0,\ldots,9$$

The denoising-variance signal

Record clean-action estimates before the Euler update. For I={5,6,7,8,9}, L=5, future action j=0,...,49 and valid coordinate d=0,...,6, use population variance:

$$Z_i = X^{(i)} - t_i v_i,\ \mu_{j,d} = (1/L)\ \Sigma_{i\in I}\ Z_{i,j,d}$$
$$V_j = \Sigma_{d=0..6}\ (1/L)\ \Sigma_{i\in I}\ (Z_{i,j,d} - \mu_{j,d})^2$$

V measures disagreement about the same future action within one generation. It uses normalized valid channels, excludes padding and reduces saved float32 estimates in float64. This is a proxy for internal stability, not calibrated failure risk or variance across successive replanning calls.

Threshold, original rule and execution cap

W contains every value from all 50 positions in each of up to five preceding calls. Repeated values remain. Add the current V only after its decision; reset history every episode.

$$\tau = \text{mean}(W) + 2\ \text{std}_{pop}(W);\ j^* = \text{first } j \text{ with } V_j > \tau$$
$$K_{raw} = 50 \text{ if no crossing; otherwise } \max(1, j^*)$$
$$K = \min(C, K_{raw})$$

The first call uses K=1 and no threshold. Execute the original final X(10) prefix, never the mean intermediate estimate. A crossing at j=4 permits indices 0-3; the minimum of one action can force execution when j=0 crosses. The cap applies to both crossing and no-crossing branches.

C=50 preserves the uncapped rule. C=1 always replans after one action, duplicating fixed K=1 under identical inputs and sampling, so it is not collected as a separate treatment. The cap does not shorten H or reduce M; its purpose is to limit open-loop execution.

A matched comparison using retained controls

Three tasks, ten initial states per task (0-9), and policy seed 11 form 30 matched blocks. Each block contains fixed K=1,5,10,25,50 and DVAC C=5,10,25,50: 270 episodes in total. The existing 150 fixed-control episodes are reused; only 120 capped episodes are newly collected.

Task	LIBERO suite	Action limit	Starts / method
Soup into basket	libero_object	280	10
Bowl onto plate	libero_spatial	280	10
Open middle drawer	libero_goal	300	10

Controlled configuration

Checkpoint and backbone revisions are pinned. Every method uses H=50, M=10, FP32, 20 Hz simulated control, ten settling steps and the same sampling-seed rule. All new adaptive caps share L=5, alpha=2, five-call history, minimum K=1 and first-call K=1 initialization. The latter initialization and seven-channel mask are explicit SmolVLA adaptation choices.

The 30 new task/state blocks and four methods within each block are shuffled with seed 20261024, then run sequentially on one RTX 2060. The cap variants have identical resource limits and diagnostic instrumentation. Hyperparameters and comparison rules were fixed before these runs.

Compatibility checks

Check	Coverage	Required result
Pinned sources and environment	Both historical control roots	Matching checkpoint, base code, software, assets and task limits
Plain versus instrumented inference	Three excluded task inputs	Every final 50 x 7 action value exactly equal
Historical reset / first prediction	All 30 matched situations	Identical reset hashes, noise seed and complete first chunk
Collection-time gate	Each of 120 capped episodes	The same reset and first-chunk check before any robot action

Instrumentation check	Input	Maximum error
Soup into basket	State 21 / seed 101	0
Bowl onto plate	State 21 / seed 101	0
Open middle drawer	State 21 / seed 101	0

The separate historical check generated one chunk per situation and executed no robot actions. Exact equality supports compatibility on the checked inputs; it does not prove equivalence on every future observation. All original raw controls and their saved evidence checksums remain intact.

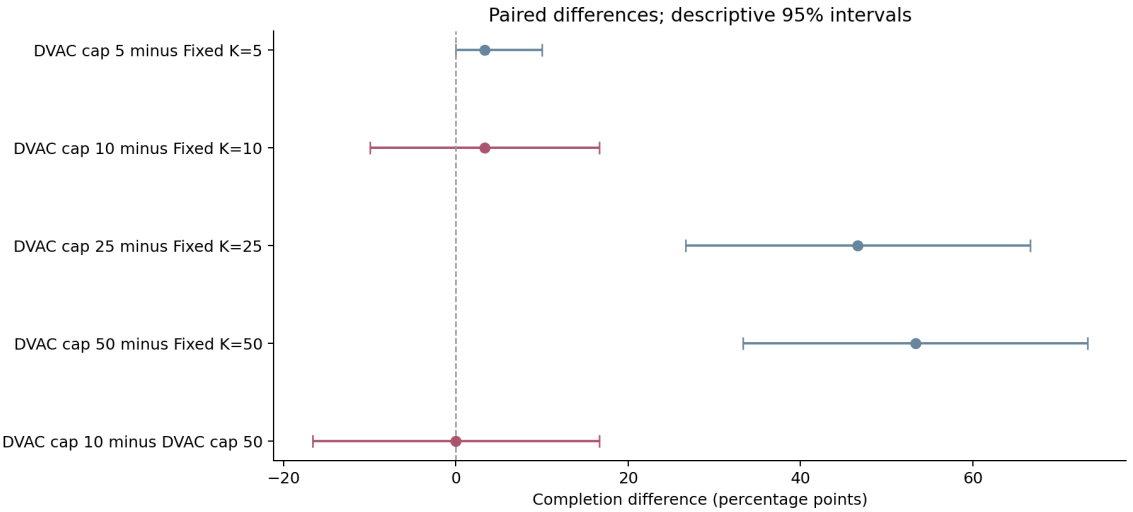
Timing boundary

Historical controls used six Torch CPU threads and did not collect internal FM diagnostics; new adaptive variants use one thread and record those diagnostics. Therefore no fixed-versus-adaptive wall-time or inference-speed claim is made. Model calls are the common cost measure. Page 6 reports timings only among the four newly collected adaptive caps.

Paired differences: does the cap help?

Two comparisons were designated: A10 versus F10 tests whether earlier updates help under the same maximum interval; A10 versus A50 tests the effect of restricting long prefixes. A denotes adaptive cap C and F denotes fixed K. Every cap-versus-matching-fixed result is also shown.

Comparison	Completion difference, pp [95% interval]	Call difference / episode [95% interval]	Only first / only second succeeds
A5 - F5	+3.3 [+0.0, +10.0]	+8.9 [+5.9, +11.8]	1 / 0
A10 - F10	+3.3 [-10.0, +16.7]	+10.9 [+6.9, +14.9]	3 / 2
A25 - F25	+46.7 [+26.7, +66.7]	+4.5 [+2.2, +7.0]	15 / 1
A50 - F50	+53.3 [+33.3, +73.3]	+4.4 [+2.5, +6.4]	17 / 1
A10 - A50	+0.0 [-16.7, +16.7]	+16.9 [+12.9, +20.9]	3 / 3



Differences are first method minus second on the same starts. Discordant counts show where the methods disagree; shared successes and shared failures remain in the complete matrix on page 7. The complete 36 pairwise comparisons are retained in paired_contrasts.csv.

How uncertainty is estimated

Each of 10,000 bootstrap draws samples ten initial states with replacement within each task, keeping all nine methods together. Task means receive equal weight. Ratios are recomputed in each draw; 2.5th and 97.5th percentiles form the displayed intervals. The random seed is 20261024. The experimental unit is a physical task/initial-state setting.

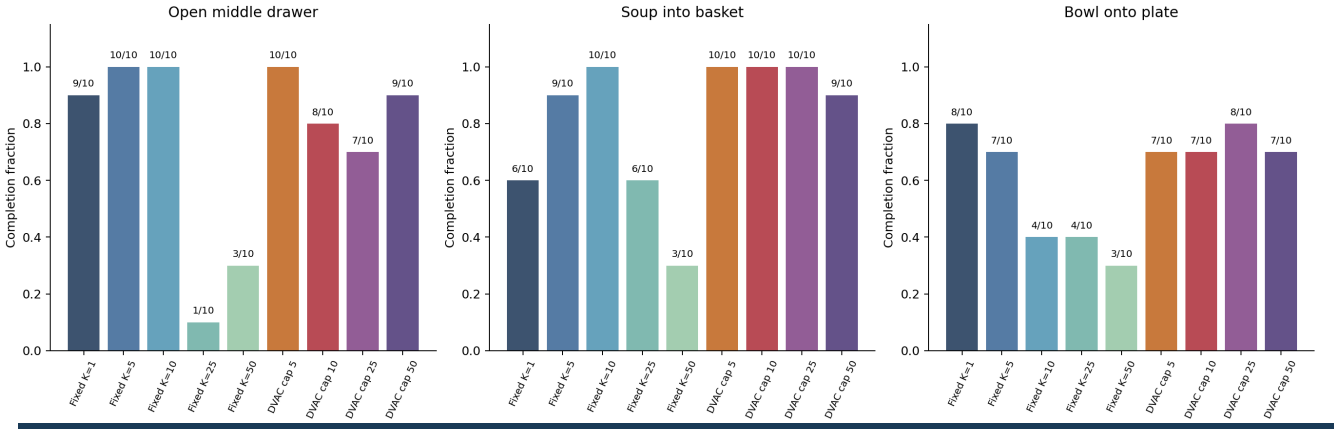
These are descriptive, exploratory intervals on previously examined states. They are not multiplicity-adjusted confirmatory tests. An interval containing zero does not show equivalence; one excluding zero does not remove selection or generalization limits. When every observed state has the same outcome, a bootstrap interval can collapse without guaranteeing future success.

Calls and episode length must be read together

For a realized episode of T actions, a cap-C schedule needs at least ceil(T/C) calls. It cannot use longer prefixes than fixed K=C. However, the methods may reach success at different times, so calls per episode and calls per 100 executed actions answer different questions.

Task-specific behavior and motion diagnostics

Each method sees the same ten initial states for each task. The three manipulation behaviors can respond differently to an execution cap; an aggregate mean can hide those differences.



Task	F1	F5	F10	F25	F50	A5	A10	A25	A50
Soup into basket	6/10	9/10	10/10	6/10	3/10	10/10	10/10	10/10	9/10
Bowl onto plate	8/10	7/10	4/10	4/10	3/10	7/10	7/10	8/10	7/10
Open middle drawer	9/10	10/10	10/10	1/10	3/10	10/10	8/10	7/10	9/10

F1-F50 are fixed intervals; A5-A50 are adaptive maximum intervals. Exact per-task estimates, call counts and descriptive intervals are available in by_task.csv.

Measured trajectories and gripper commands

Method	Mean actions / episode	Mean end-effector path, m	Mean close-command events
Fixed K=1	162.0	0.878	9.40
Fixed K=5	154.9	0.981	4.30
Fixed K=10	151.7	1.063	5.63
Fixed K=25	229.4	1.383	4.57
Fixed K=50	239.9	1.286	3.40
DVAC cap 5	137.6	0.974	2.83
DVAC cap 10	150.4	0.949	2.53
DVAC cap 25	157.2	1.060	1.57
DVAC cap 50	138.5	0.994	1.50

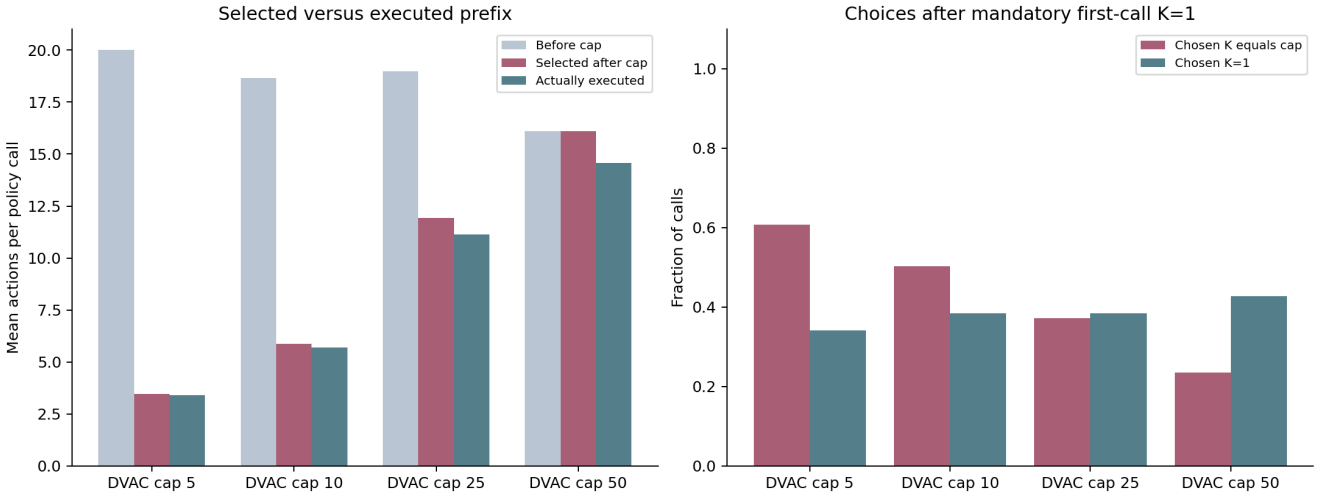
Path length sums the end-effector displacement after executed actions. A close-command event is a transition to a positive gripper command; it is not a successful grasp. Longer paths or repeated closing may reflect retries, task progress or an unsuccessful trajectory. Early termination also changes these totals.

Replanning changes the action chunk after a new observation. That can enable a correction, but a changed command alone does not prove that an error was corrected. Failure mechanisms need video/contact evidence. This report does not convert command counts into grasp-error rates.

Success uses the native LIBERO predicate and stops the episode immediately. It may occur before an object is released; post-release stability is not evaluated.

How often did the cap actually bind?

The raw DVAC choice is computed over all 50 predicted actions, then clipped by C. A cap binds only when this clipping changes K. Selected K and the actually executed prefix differ when success or termination interrupts a plan.



Cap C	Mean raw K	Mean chosen K	Mean executed	Calls limited by cap	K=1 after first call
5	20.02	3.45	3.42	57.9%	34.2%
10	18.65	5.87	5.70	47.0%	38.4%
25	18.97	11.92	11.15	34.3%	38.4%
50	16.09	16.09	14.58	0.0%	42.7%

Mean K weights policy calls equally. Cap-binding fractions include first calls; the final column excludes the mandatory first-call K=1 initialization. Different caps create different trajectories and observations, so their raw-K distributions are not evaluations of one shared variance sequence.

Calls without a variance crossing after initialization: C=5: 261/1177; C=10: 153/761; C=25: 100/393; C=50: 60/255. These raw choices are still limited by C.

Measured runtime among adaptive caps only

Cap C	Inference s / call	Selector ms / call	Compute s / episode	Wall s / episode
5	0.790	0.060	35.73	40.13
10	0.790	0.058	25.07	30.07
25	0.792	0.056	15.58	20.09
50	0.799	0.051	11.47	15.73

Inference includes the instrumented policy prediction and variance calculation. Selector time is measured separately. Compute time is inference + selection + simulator stepping. Episode wall time includes reset, rollout and video closure, excluding JSON serialization, hashing and publication of the evidence directory.

The synchronous simulator pauses during inference. Playback follows simulation time and does not show compute pauses as physical motion. These measurements do not establish real-time control performance, asynchronous latency tolerance, or fixed-versus-adaptive speedup.

Every matched outcome and call count

Each cell is outcome / model calls: S = success, F = failure. All rows use policy seed 11. F1,F5,F10,F25,F50 denote fixed K; A5,A10,A25,A50 denote adaptive cap C. Every method has exactly one episode in every row.

Task	State	F1	F5	F10	F25	F50	A5	A10	A25	A50
Soup	0	F/280	S/31	S/14	S/5	S/3	S/35	S/21	S/14	S/8
Soup	1	F/280	S/29	S/14	F/12	F/6	S/32	S/21	S/7	S/5
Soup	2	S/162	S/29	S/14	S/6	F/6	S/30	S/17	S/12	S/5
Soup	3	F/280	S/30	S/14	F/12	F/6	S/35	S/20	S/8	F/14
Soup	4	S/164	S/30	S/14	F/12	S/3	S/36	S/24	S/10	S/16
Soup	5	F/280	S/30	S/14	S/5	F/6	S/35	S/23	S/6	S/8
Soup	6	S/162	S/29	S/14	S/5	F/6	S/33	S/17	S/9	S/5
Soup	7	S/120	F/56	S/14	F/12	F/6	S/37	S/17	S/9	S/6
Soup	8	S/150	S/27	S/14	S/5	S/3	S/37	S/17	S/13	S/12
Soup	9	S/146	S/30	S/14	S/5	F/6	S/39	S/17	S/7	S/6
Bowl	0	S/75	S/15	F/28	F/12	F/6	S/22	F/48	S/11	S/12
Bowl	1	F/280	F/56	F/28	F/12	S/4	F/72	F/44	F/27	F/23
Bowl	2	F/280	F/56	F/28	F/12	F/6	F/77	F/41	S/16	F/20
Bowl	3	S/80	S/19	S/8	S/4	F/6	S/22	S/14	S/7	F/23
Bowl	4	S/70	S/15	S/8	F/12	F/6	S/20	S/13	S/8	S/6
Bowl	5	S/99	S/17	F/28	S/7	S/3	S/20	S/13	S/13	S/6
Bowl	6	S/70	S/15	S/8	F/12	S/2	S/18	S/12	S/9	S/7
Bowl	7	S/78	F/56	S/13	S/4	F/6	F/66	S/12	S/8	S/7
Bowl	8	S/80	S/17	F/28	F/12	F/6	S/32	S/10	S/6	S/5
Bowl	9	S/77	S/17	F/28	S/4	F/6	S/21	S/14	F/25	S/8
Drawer	0	F/300	S/48	S/12	S/10	F/6	S/50	F/66	S/28	S/7
Drawer	1	S/154	S/26	S/11	F/12	F/6	S/48	S/25	S/11	S/7
Drawer	2	S/141	S/23	S/12	F/12	F/6	S/44	S/34	S/10	S/7
Drawer	3	S/153	S/51	S/12	F/12	S/3	S/51	F/57	S/15	S/7
Drawer	4	S/188	S/25	S/13	F/12	F/6	S/59	S/27	S/24	S/7
Drawer	5	S/138	S/33	S/12	F/12	S/3	S/51	S/27	F/25	S/6
Drawer	6	S/161	S/51	S/13	F/12	F/6	S/51	S/30	S/22	F/21
Drawer	7	S/146	S/28	S/12	F/12	S/3	S/51	S/54	F/25	S/7
Drawer	8	S/143	S/28	S/12	F/12	F/6	S/43	S/31	S/13	S/7
Drawer	9	S/124	S/23	S/11	F/12	F/6	S/40	S/25	F/25	S/7

All 270 outcomes are shown. The underlying numerical matrix is retained in outcome_matrix.csv; the color-coded overview is outcome_matrix.png. These complete records prevent favorable examples from substituting for the overall comparison.

Predetermined demonstration

The main recording uses state 0 and seed 11 for all three tasks, showing fixed K=10 beside DVAC C=10 and C=50. Instructions, robot motion, replanning markers, chosen K, cap, call count and outcome are displayed. Every preselected task is included regardless of outcome.

Reviewed illustrations of outcome disagreements

These illustrations are selected using the outcomes and are not an unbiased error sample. Observed motion is separated from hypotheses about its cause. The complete outcome matrix remains the basis for comparing methods.

Transport continues without the can

Soup into basket | state 3 | policy seed 11



Observed: At steps 59-100, fixed K=10 and cap 10 carry the can above the table; with cap 50 it stays at its starting location while the empty gripper moves toward the basket. Fixed K=10 and cap 10 finish at steps 139 and 143. Cap 50 does not recover in the inspected frames and times out at step 280.

Hypothesis: The cap-50 sequence is consistent with an unsuccessful pickup followed by continued transport motion. The images do not identify the contact error or show that a particular extra observation would have prevented it.

Frequent updates do not resolve every failure

Bowl onto plate | state 0 | policy seed 11



Observed: Cap 50 moves the bowl toward the plate at steps 55-81 and succeeds at step 81. Fixed K=10 and cap 10 repeatedly reposition nearby and time out at step 280; their plates visibly tilt (fixed: 77-81; cap 10: 90-97). Gripper-command sign changes are not counted as grasp attempts.

Hypothesis: More frequent updates did not resolve these two failed trajectories. Continued resampling may interrupt a useful motion sequence, but these videos cannot separate that effect from earlier pose and action differences.

Interpretation, limits and reproducibility

For the designated A10-F10 comparison, completion changed by +3.3 percentage points and calls by +10.9 per episode. The sample-mean frontier contains Fixed K=5, Fixed K=50, DVAC cap 5, DVAC cap 50: no other method matches or improves both completion and call count while strictly improving one. This describes the observed trade-off, not a universally optimal rule.

What the experiment cannot establish

The fixed outcomes were already examined and informed the choice to study caps. This is an exploratory comparison on seen states, not a held-out confirmation. Only one policy seed and three fixed tasks are included. The intervals describe variation across these initial situations; they do not estimate uncertainty over arbitrary new tasks or robustness to generation noise.

A stable internal flow trajectory can still lead to an inaccurate action. More frequent replanning cannot guarantee recovery from a missed grasp or systematic model error. Changing K combines new observations and stochastic regeneration; this design does not isolate their causal effects. Video illustrations can suggest mechanisms but do not provide contact-mechanical proof or error rates.

This comparison evaluates the full DVAC rule and its cap. Isolating the contribution of the variance signal would require a control schedule matched on model-call budget or execution-interval distribution. Better completion alone would not establish that V predicts errors.

Reproduce the retained-control comparison

```
bash scripts/collect_capped.sh check configs/capped.json \
 --output outputs/capped-check/invariance.json
bash scripts/collect_capped.sh reference-check configs/capped.json \
 --reference-roots outputs/compact outputs/validation \
 --output outputs/capped-check/historical-reference.json
bash scripts/collect_capped.sh --config configs/capped.json --output outputs/capped \
 --reference-roots outputs/compact outputs/validation
python scripts/analyze_capped.py outputs/capped outputs/capped-report --decode-videos
python scripts/build_capped_report.py --discussion outputs/capped-report/discussion.json
```

Use the pinned Linux/CUDA runtime for collection, with the documented CPU, memory and thread limits. Analysis/PDF tools use the artifact environment. The report rebuild verifies every episode and recomputes the paired bootstrap. The adjacent JSON binds all used statistics, figures, raw evidence and source files to hashes. Render and inspect all PDF pages before delivery.

```
Checkpoint: HuggingFaceVLA/smolvla_libero
Revision: 6721902bc4d61e50a3bfdb11dfb4cb626f05d102
Statistics SHA-256: eb164137a9efdaa89e1e8fcca4ae86ae2f37a6ba97b5a5b2c223f84cb9101a6a
```

References

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2. SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics.
3. Pinned SmolVLA LIBERO checkpoint.
4. LeRobot 0.6.1 SmolVLA implementation.